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Q.3 (a) What is a priority queue? How would you implement it.

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(b) How Post-order traversal works on a binary tree. Write a non-recursive procedure for the same.

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(c) What is STL? What are its key components and features? Write code to demonstrate use of vector class.

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Q.4 (a) What is a binary heap? How could you implement it.

[6]

(b) Write an algorithm/code to generate a magic square. (Note: A magic square is a square matrix whose sum of elements row-wise, column-wise and diagonal is same.)

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(c) What is an AVL Tree? How would final AVL tree look like when you enter months of a year in reverse order. (First, you need to insert December then November...)

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Q.5 (a) What is Hashing? Why it is considered as an efficient data structures? Discuss any one hashing function.

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(b) What is Binary Threaded Tree? What is its purpose? How would you implement it.

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(c) Write an algorithm/procedure for implementing Depth First Search.
